Moments and Probability Code

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MATH 241

Week 10, Lecture #1 — Ch. 5.9, 6.1-6.2, 6.4

Last Time and Today

- Continuous RVs
  - $E(X)$
  - Uniform, normal, and exponential distributions
- Today: Quantiles, moments, and R code

Reminders

1. HW7 due Monday, 11/9 at 11:59pm
2. Exam #2 next Tuesday, 11/10 during class
   - More on this at the end of class!

Summary

	Discrete	Continuous
CDF ($F_X(x)$)	$\sum_{a \leq x} P(X = a)$	$\int_{-\infty}^{x} f_X(t)dt$
PDF ($f_X(x)$)	$P(X = x)$	$F'_X(x)$
$E(X)$	$\sum_x x\, P(X = x)$	$\int_{-\infty}^{\infty} x\, f_X(x)dx$
LOTUS ($E(g(X))$)	$\sum_x g(x)\, P(X = x)$	$\int_{-\infty}^{\infty} g(x)\, f_X(x)dx$
Common Dists.	Bin(n, p)	N(μ, σ^2)
	Geom(p)	Exp(λ)
	Pois(λ)	Unif(a, b)

Table: Summary of Discrete and Continuous RVs

Warm-Up: Identify the Distribution

For each of the following situations, I want you to:

- Identify the common distribution of X and determine what the population parameters are.

Here are the situations:

1. Adult heights in the state of New York.
2. The number of attempts it takes for a former Knicks player to make their first free throw.
3. Selecting a completely random time between noon and one o'clock for a brief phone call.

R Code

Say we want to find items for $X \sim N(\mu = 4, \sigma^2 = 2)$.

- `pnorm(x, mu, sigma)`: lower-tailed probability (i.e. $P(X \leq x)$)
- `dnorm(x, mu, sigma)`: value of $f_X(x)$
- `qnorm(q, mu, sigma)`: value of x for the $(100 \times q)$th percentile (i.e. x such that $P(X \leq x) = q$)

For the first arguments: x will always go first in your d and p functions; q will always go first in your q function.

R Code

Say we want to find items for $X \sim N(\mu = 4, \sigma^2 = 2)$.

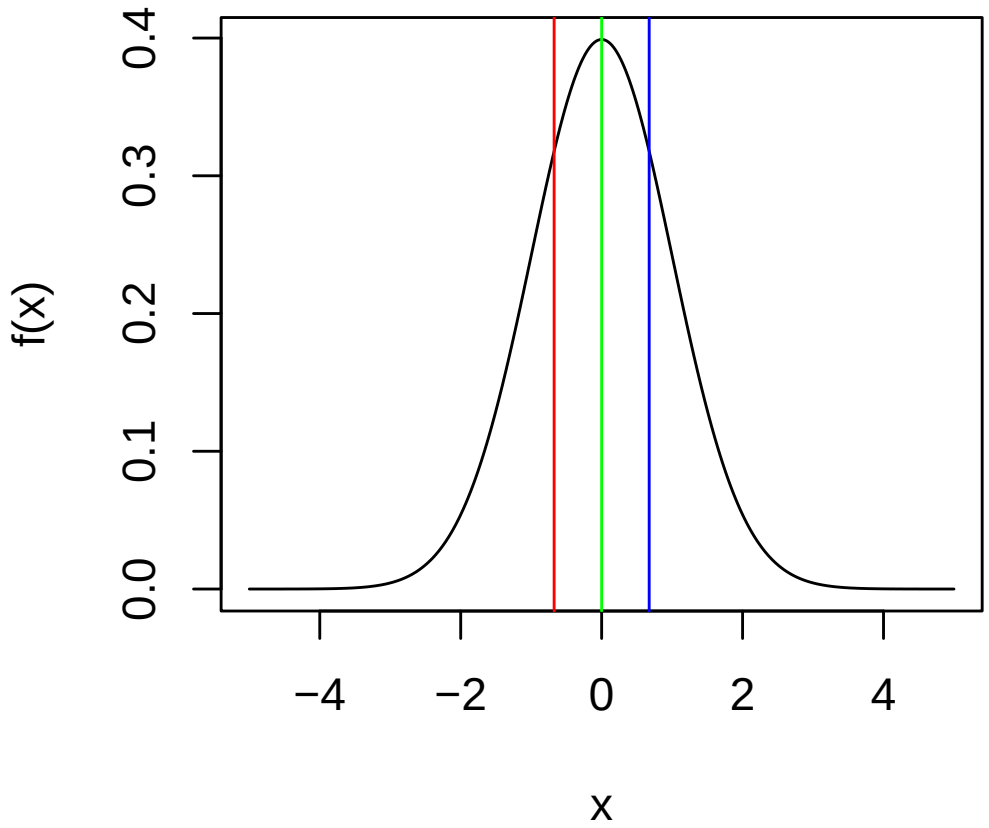
- `pnorm(0, 4, sqrt(2))`: lower-tailed probability (i.e. $P(X \leq 0)$)
- `dnorm(0, 4, sqrt(2))`: value of $f_X(0)$
- `qnorm(0.5, 4, sqrt(2))`: value of x for the 50th percentile (i.e. x such that $P(X \leq x) = 0.5$)

For the first arguments: x will always go first in your d and p functions; q will always go first in your q function.

R Code for Different Distributions

Dist.	Code	2nd Argument	3rd Argument
$N(\mu, \sigma^2)$	`_norm()`	$mean = \mu$	$sd = \sqrt{\sigma^2}$
$Unif(a, b)$	`_unif()`	$min = a$	$max = b$
$Exp(\lambda)$	`_exp()`	$rate = \lambda$	
$Bin(n, p)$	`_binom()`	$size = n$	$prob = p$
$Geom(p)$	`_geom()`	$prob = p$	
$Pois(\lambda)$	`_pois()`	$lambda = \lambda$	

Summary Statistics for a Distribution



Percentiles and Median

c is the $100 \times q$th percentile of a continuous random variable X if:

$$P(X \leq c) = q$$

Definition (6.1.1 — Median)

c is a median of a random variable X if:

$$P(X \leq c) \geq 0.5$$
$$P(X \geq c) \leq 0.5$$

Examples

Say we want to find the first quartile, median, and third quartile of the waiting times at a bank, which follows an exponential distribution with $\lambda = \frac{1}{3}$.

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(1) Find $F_X(x)$:

$$F_X(x) = 1 - e^{-\frac{1}{3}x} \quad \text{for } x \geq 0$$

(2) Let $F_X(x) = q$, solve for x:

$$\begin{aligned} q &= 1 - e^{-\frac{1}{3}x} \\ \implies q - 1 &= -e^{-\frac{1}{3}x} \\ \implies 1 - q &= e^{-\frac{1}{3}x} \\ \implies log\{1 - q\} &= -\frac{1}{3}x \\ \implies -3log\{1 - q\} &= x \end{aligned}$$

Examples

Say we want to find the first quartile, median, and third quartile of the waiting times at a bank, which follows an exponential distribution with $\lambda = \frac{1}{3}$.

(3) Substitute $q = 0.25, 0.5, 0.75$ to find x:

$$x_{0.25} = -3log\{1 - 0.25\} \approx 0.8630$$
$$x_{0.5} = -3log\{1 - 0.5\} \approx 2.0794$$
$$x_{0.75} = -3log\{1 - 0.75\} \approx 4.1589$$

Mode

Definition (6.1.2 — Mode)

For a discrete RV X, we say that c is a mode of X if it maximizes the PMF, i.e.:

$$P(X = c) \geq P(X = x) \qquad \text{for all } x$$

Recall: Moments

The nth moment of X can be defined by:

$$E(X^n)$$

- e.g. $E(X^2)$ to find the variance
- There are different types of moments

Three Types of Moments

Raw	Central	Standardized
$E(X^n)$	$E((X-\mu)^n)$	$E\left(\frac{X-\mu}{\sigma}\right)^n$

Examples:

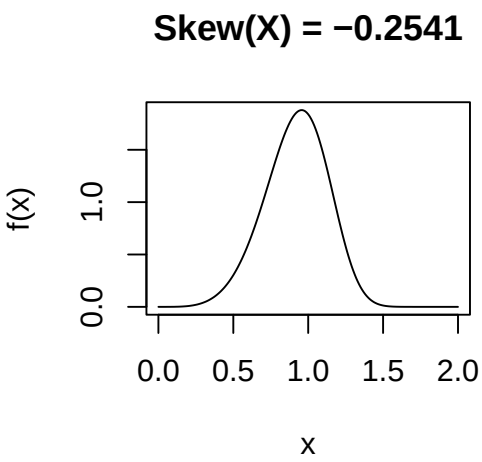
- Mean: *first raw moment*
- Variance: *second central moment*
- Skewness: *third standardized moment*
- Kurtosis: *fourth standardized moment*, shifted to the left by 3
  - $E\left(\frac{X-\mu}{\sigma}\right)^4 - 3$

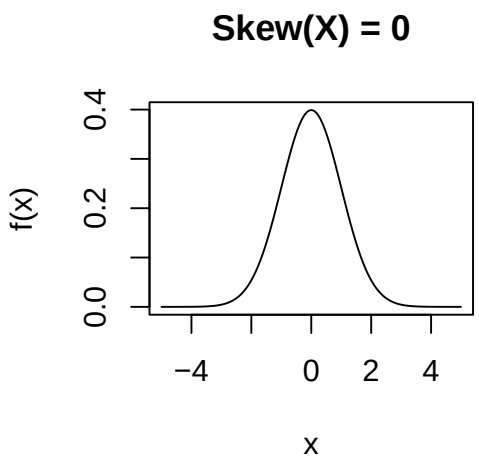
Skewness

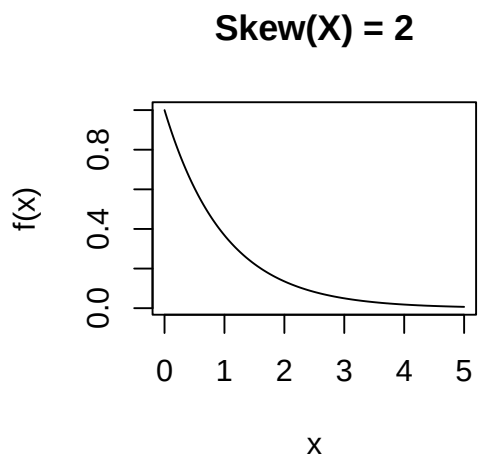
$$Skew(X) = E\left(\frac{X-\mu}{\sigma}\right)^3$$

- Third standardized moment
- Measures how data is spread around the center of our distribution
- If...
  - Left-skewed: $Skew(X) < 0$
  - Symmetric: $Skew(X) = 0$
  - Right-skewed: $Skew(X) > 0$

Examples


Skew(X) = −0.2541
f(x)
0.0
1.0
0.0
0.5
1.0
1.5
2.0
x


Skew(X) = 0
f(x)
0.0
0.2
0.4
−4
0
2
4
x

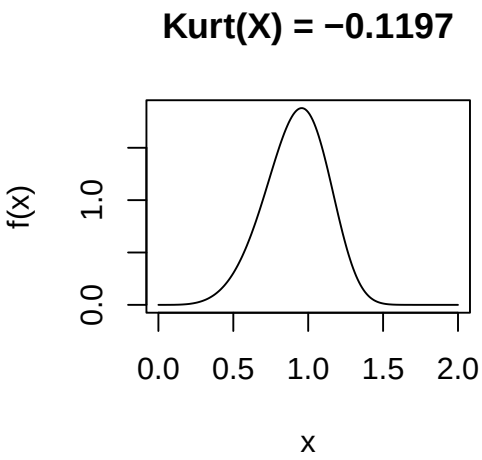

Skew(X) = 2
f(x)
0.0
0.4
0.8
0
1
2
3
4
5
x

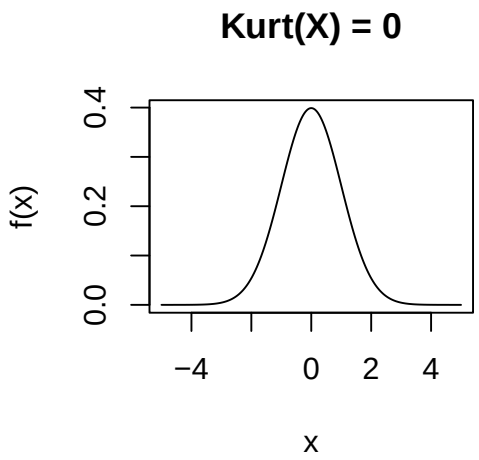
Kurtosis

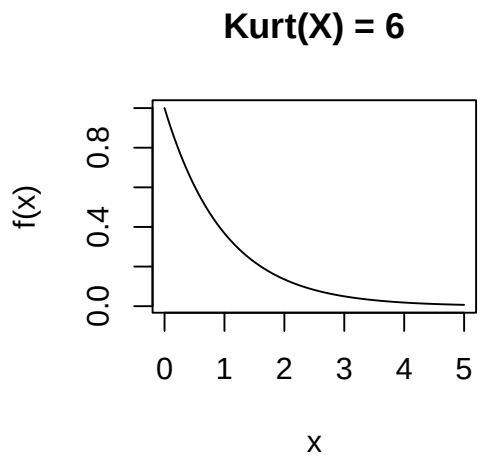
$$Kurt(X) = E\left(\frac{X-\mu}{\sigma}\right)^4 - 3$$

- Fourth standardized moment minus 3
- Measure how heavy or light the tails are (e.g. the thickness of the tails)
- If...
  - Lighter tails: $Kurt(X) < 0$
  - Normal tails: $Kurt(X) = 0$
  - Heavier tails: $Kurt(X) > 0$

Examples


Kurt(X) = −0.1197
f(x)
0.0
1.0
0.0
0.5
1.0
1.5
2.0
x


Kurt(X) = 0
f(x)
0.0
0.2
0.4
−4
0
2
4
x


Kurt(X) = 6
f(x)
0.0
0.4
0.8
0
1
2
3
4
5
x

Why Moments?

- Provides metrics on certain characteristics of a probability distribution
- Method of Moments (MoM) estimator via sample moments:

$$M_k = \frac{1}{n}\sum_{j=1}^{n} X_j^k$$

  - Useful for estimating population parameters
- *Note*: You do not need to worry about calculating these on your own – just need to identify which moments capture which distribution characteristics